MTh 623

① Define Torsion tensor?

② prove that $ds^2 = d\bar{s}^2$

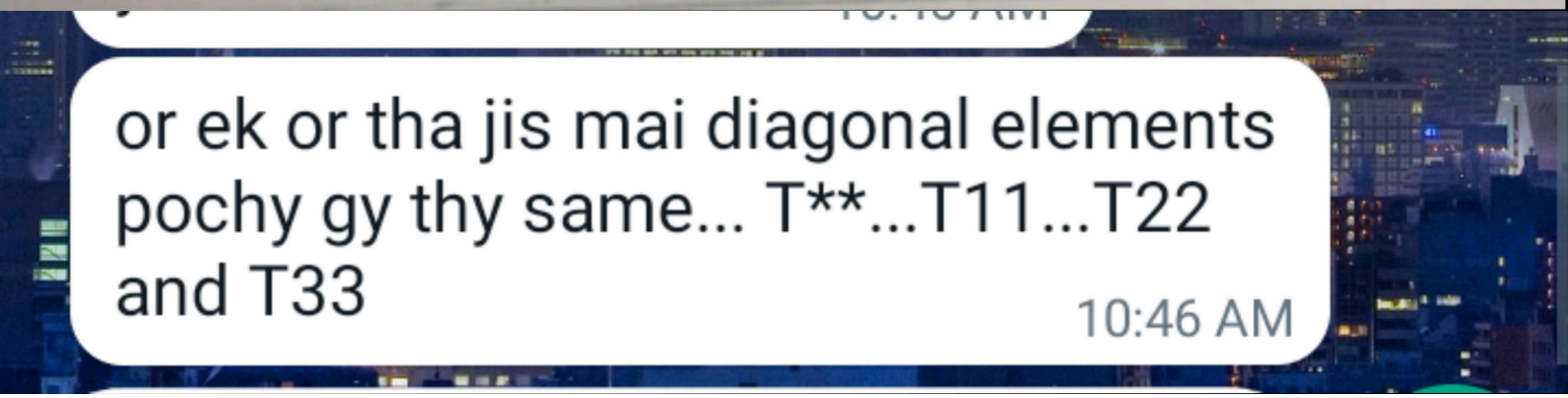
③ If $U^i = (c, v^x, 0, 0)$. that prove that $T^{ij} = \rho U^i U^j$

④ Explain Normal stress

⑤ If $\frac{\partial g_{ij}}{\partial y_k} = 0$ then prove that $\frac{\partial g^{ij}}{\partial y^k} = 0$

⑥ $g_{\alpha\beta,\gamma} = \delta^{\gamma}_{\alpha}$ is ko prove krnat tha but sai say gad nai ha.

Ziada ter paper last topic say aya tha. MCQs last say thay.



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Riemannian Curvature:

The Riemannian or sectional curvature relative to a given metric (g_{ij}), a Riemannian manifold and two linearly independent tangent vectors at the same point, (v^i) and (v^i) is defined as:

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The Components of Energy Momentum Tensor of Dust in Minkowski spacetime (Flat space)

Here we discuss two cases: when frame is comoving and non comoving.

Case I:

A frame moving with the flow of the dust fluid is called comoving frame. For a co-moving observer, the four velocity reduces to

$$u^i = (c, 0, 0, 0)$$

In this case, the stress-energy tensor takes on the remarkably simple form

$$T^{ij} = \rho u^i u^j \rightarrow T^{00} = \rho u^0 u^0 = \rho c^2$$

$$T^{ij} = \begin{bmatrix} c^2\rho & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

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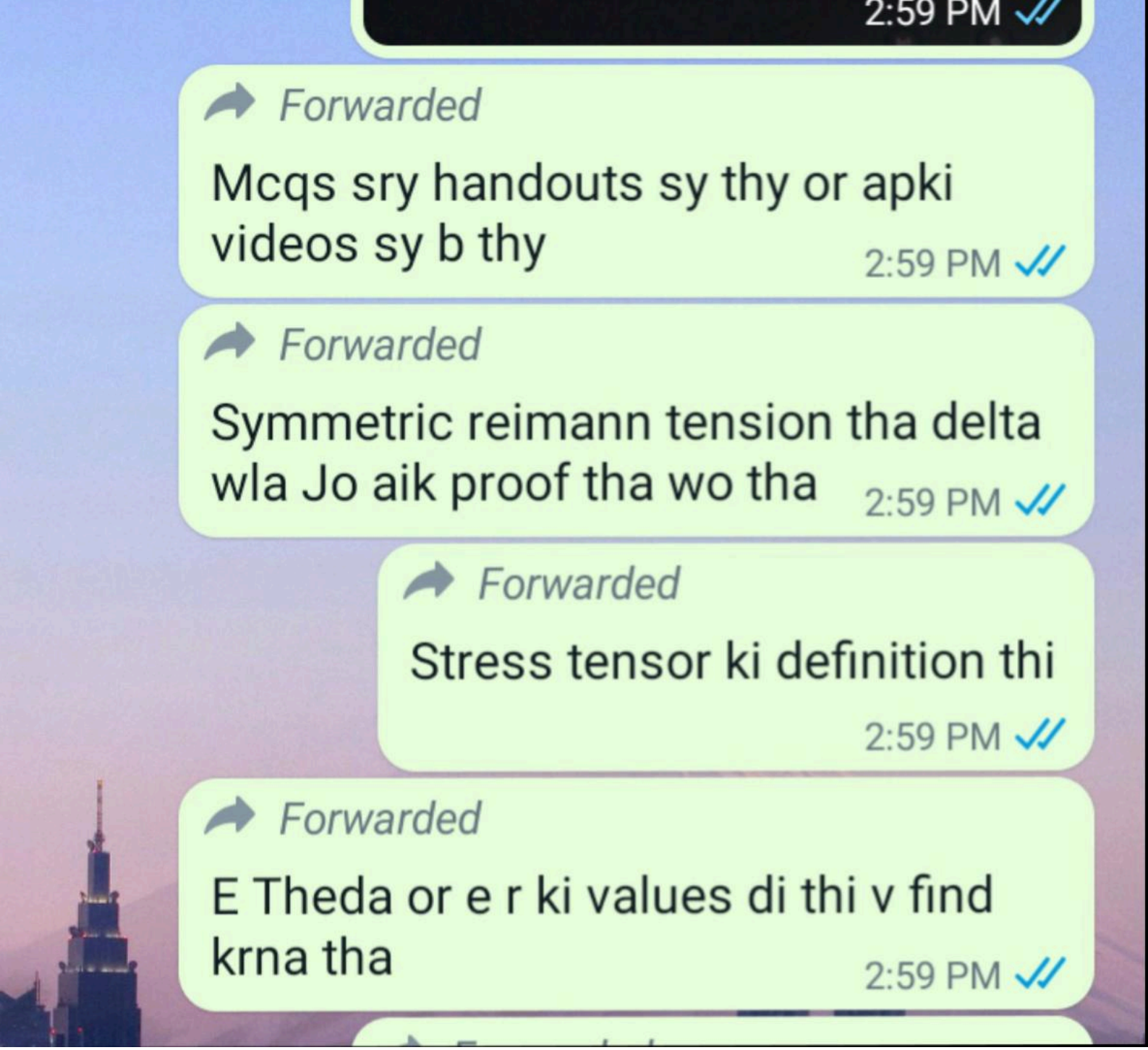
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2.Normal stress ,shear stress nikalna tha
3. Basis vectors deay thy wo vector my put kr k partially differentiate krna tha ek long or tha yad nae
Short my torsion tensor definition
Interval batana tha k light like hai ya nae
Christoffel symbol ek nikalna tha polar coordinates ka
Ek koe or tha wo yad nae

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Question: Compute the components of the Riemann tensor for a unit 2-sphere, where

$$ds^2 = d\theta^2 + \sin^2\theta\, d\phi^2$$

and the non-zero christoffel symbols are:

$$\Gamma^1_{22} = -\sin\theta\cos\theta$$

$$\Gamma^{12}_{21} = \Gamma^2_{12} = \cot\theta$$

$x^1, x^2, x^3 \to r, \theta, \phi$
but $r = 1$
$\Rightarrow x^1, x^2 \to \theta, \phi$

Solution:
Using the fact that the non-zero components of the Riemann tensor in two dimensions

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Question: Find $\nabla^2\phi$ for $\phi = \frac{x^1 x^2}{2}$ in polar coordinates.

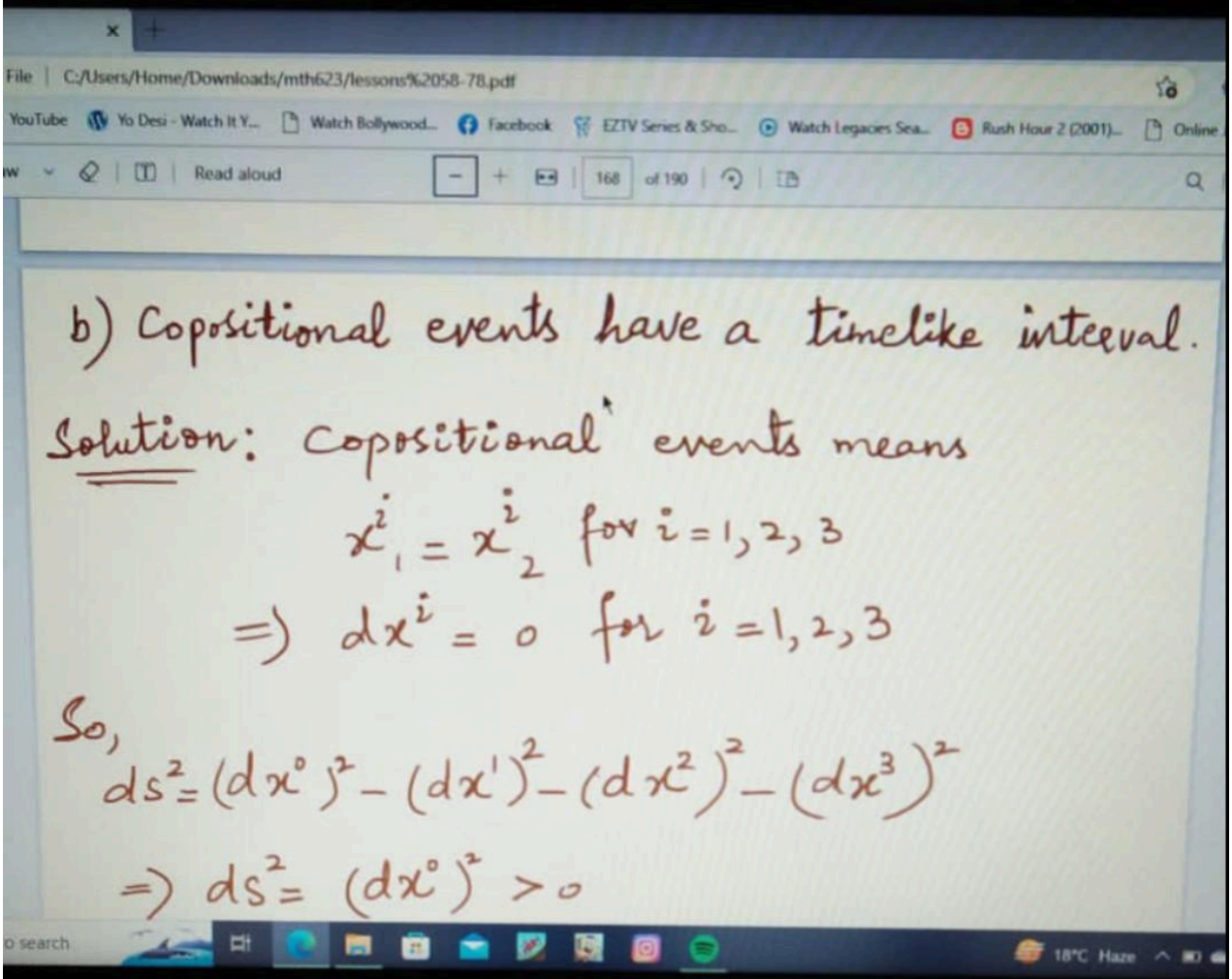
Solution: First consider:

$$\nabla\phi = \frac{\partial\phi}{\partial x^i}\hat{e}_i = \frac{\partial\phi}{\partial x^1}\hat{e}_1 + \frac{\partial\phi}{\partial x^2}\hat{e}_2$$

$$\Rightarrow \nabla\phi = \frac{\partial}{\partial x^1}\left(\frac{x^1 x^2}{2}\right)\hat{e}_1 + \frac{\partial}{\partial x^2}\left(\frac{x^1 x^2}{2}\right)\hat{e}_2$$

$$\nabla\phi = \frac{x^2}{2}\hat{e}_1 + \frac{x^1}{2}\hat{e}_2$$

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b) Coposition al events have a timelike inteeval.
Solution: Coposition al events means
$x^i_1 = x^i_2$ for $i = 1, 2, 3$
$\Rightarrow dx^i = 0$ for $i = 1, 2, 3$
So,
$ds^2 = (dx^0)^2 - (dx^1)^2 - (dx^2)^2 - (dx^3)^2$
$\Rightarrow ds^2 = (dx^0)^2 > 0$
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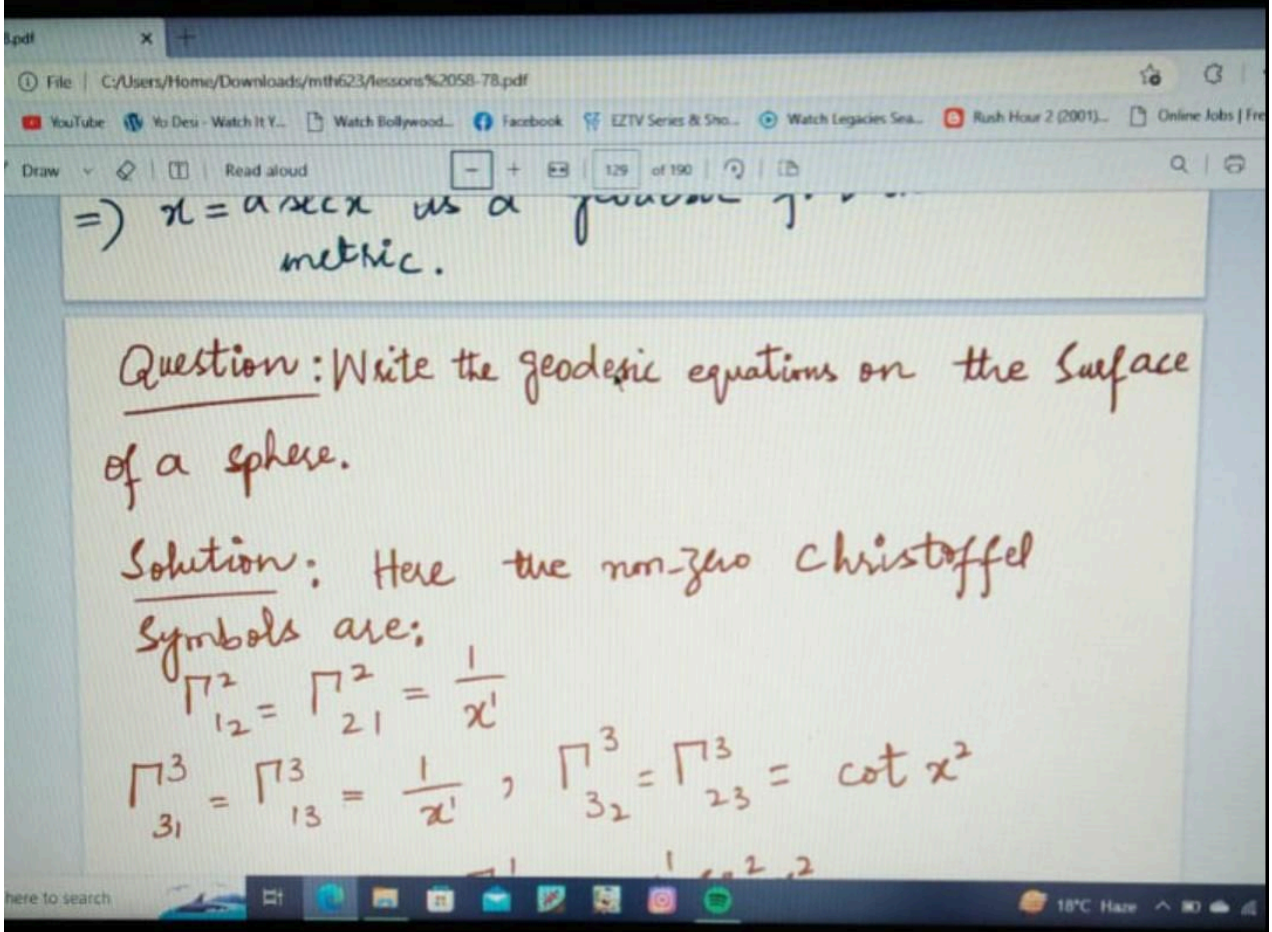
Light Cone: A flash of light at position $(0,0,0)$ and time $t=0$ sends out an expanding spherical wave front, with equation

$$x^2+y^2+z^2=c^2t^2 \quad \text{or}$$

$$(x^0)^2-(x^1)^2-(x^2)^2-(x^3)^2=0$$

this is the equation of the light cone in event space, relative to the inertial

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=) $x = a\sec x$ is a
metric.
Question: Write the geodesic equations on the surface of a sphere.
Solution: Here the non-zero Christoffel symbols are:
$\Gamma^2_{12} = \Gamma^2_{21} = \frac{1}{x^1}$
$\Gamma^3_{31} = \Gamma^3_{13} = \frac{1}{x^1}$, $\Gamma^3_{32} = \Gamma^3_{23} = \cot x^2$
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Question: Consider polar coordinates and find the covariant derivative $V^{\alpha}{}_{;\alpha}$ of $V^{\alpha} = r^2 \cos\theta \, \hat{e}_r - \sin\theta \, \hat{e}_\theta$.

Solution: Using the formula we have:

$$V^{\alpha}{}_{;\alpha} = V'_{;1} + V^2{}_{;2}$$

$$V^{\alpha}{}_{;\alpha} = V^{\alpha}{}_{,\alpha} + \Gamma^{\alpha}_{\beta\alpha} V^{\beta}$$

$$V'_{;1} = V'_{,1} + \Gamma^{1}_{\beta 1} V^{\beta}$$

$$V^2 \quad V^2 \quad + \Gamma^2 \quad V^{\beta}$$

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Question: Calculate the Christoffel Symbols of the second kind for the Euclidean metric in polar coordinates.

$$[g_{\mu\nu}] = \begin{bmatrix} 1 & 0 \\ 0 & (x^1)^2 \end{bmatrix} \qquad \begin{pmatrix} x^1 \to r \\ x^2 \to \theta \end{pmatrix}$$

Solution:

Using the formula:

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Question: Compute the Christoffel symbols corresponding to the Euclidean metric for spherical coordinates.

$$[g_{\mu\nu}] = \begin{bmatrix} 1 & 0 & 0 \\ 0 & (x^1)^2 & 0 \\ 0 & 0 & (x^1)^2 \sin^2 x^2 \end{bmatrix}$$

Solution: The formula for christoffel symbols is:

$$\Gamma^i_{jk} = \frac{1}{2}\left[-g_{ij,k} + g_{jk,i} + g_{ki,j}\right]$$

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Question: Show that if $[g_{ij}]$ is matrix, then for all fixed subscr α and $\beta \neq \alpha$ in the range 1,2

1) $\Gamma_{\alpha\alpha\alpha} = \frac{1}{2} g_{\alpha\alpha,\alpha}$ (not summed

2) $-\Gamma_{\alpha\alpha\beta} = \Gamma_{\alpha\beta\alpha} = \Gamma_{\beta\alpha\alpha} = \frac{1}{2} g_{\alpha\alpha\beta}$

3) All remaining Christoffel Sym

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Electrodynamics

It is the branch of Physics which deals with rapidly changing electric and magnetic fields.

Here we will discuss the applications of tensors or use of tensors in the well known Maxwell field equations of electrodynamics.

Maxwell Field Equations:

Maxwell's equations are four equations that together form a complete description of the production and interrelation of electric and magnetic fields.

In the 19th century, the physicist James Clerk Maxwell based his description of electromagnetic fields on these four equations, which express experimental laws.

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Maxwell Field Equations

electric field

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\varepsilon_0}$$ ρ → charge density, ε_0 → permittivity of vacuum

$$\nabla \cdot \mathbf{B} = 0$$ Magnetic flux

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{j} + \frac{1}{c^2}\frac{\partial \mathbf{E}}{\partial t}$$ $\frac{1}{\sqrt{\mu_0 \varepsilon_0}} = c$ (speed of light)

$\mathbf{j}$: electric current; μ_0: permeability of vacuum

Maxwell Field Equations

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\varepsilon_0}$$ → Gauss' law of electricity

$$\nabla \cdot \mathbf{B} = 0$$ → Gauss' law of Magnetism

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$ → Faraday's law of Induction

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{j} + \frac{1}{c^2}\frac{\partial \mathbf{E}}{\partial t}$$ → Ampere's law

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Question: A body in equilibrium subject to the following stress state

$$\delta = (\delta_{ij}) = \begin{bmatrix} xy & 5y \\ 5y & 4x \end{bmatrix} \qquad \delta_{ij,j} + F_i = 0$$

determine body forces.

Solution: here we use the following eqs:

$$\frac{\partial \delta_{11}}{\partial x^1} + \frac{\partial \delta_{21}}{\partial x^2} = -F_1 \quad \text{and} \quad \frac{\partial \delta_{12}}{\partial x^1} + \frac{\partial \delta_{22}}{\partial x^2} = -F_2$$

$$\Rightarrow \frac{\partial}{\partial x}(xy) + \frac{\partial}{\partial y}(5y) = -F_1$$

$$\Rightarrow F_1 = -y + 5 \Rightarrow F_1 = 5-y$$

and

$$\frac{\partial}{\partial x}(5y) + \frac{\partial}{\partial y}(4x) = -F_2$$

$$\Rightarrow F_2 = -(0+0) = 0$$

$$\vec{F} = (5-y, 0)$$



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Question: For the stress matrix $(\delta_{ij}) = \begin{bmatrix} xy & 5y \\ 5y & 4x \end{bmatrix}$ write normal and shear stresses.

Solution: Normal stresses:

$$\delta_{11} = xy \,, \quad \delta_{22} = 4x$$

Shear Stresses:

$$\delta_{12} = \delta_{21} = 5y$$

Question: Find the constraint for the stress matrix $(\delta_{ij}) = \begin{bmatrix} xy & 5y \\ 5y & 4x \end{bmatrix}$ for tensile and compressive stress.

Solution:

For tensile stress: $\delta_{ii} > 0$

$\Rightarrow \delta_{11} > 0$ and $\delta_{22} > 0$

$\Rightarrow xy > 0$ and $4x > 0$

$\Rightarrow x > 0$ and $y > 0$

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A **manifold** is a space that can be curved but is locally flat. That is, near each point the space resembles Euclidean space. One example of a curved manifold is the surface of a sphere.

The **Riemann** tensor is the most common tool used to describe the curvature of a Riemannian manifold.

The Riemann curvature tensor is a tool used to describe the curvature of n-dimensional spaces (such as Riemannian manifolds) in the field of differential geometry. The Riemann tensor plays an important role in the theories of general relativity and gravity as well as the curvature of spacetime.

Contraction of the Riemann tensor produces the **Ricci tensor**. The Ricci tensor provides a way to measure the degree to which a space differs from Euclidean space.

Contraction of the Ricci tensor produces the scalar curvature or **Ricci scalar**. The Ricci scalar is the simplest curvature invariant of a manifold.

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Riemannian Curvature:

The Riemannian or Sectional curvature relates to a given metric (g_{ij}), a Riemannian manifold and two linearly independent tangent vectors at the same point, (U^i) and (V^i) it defined as:

$$K = K(\vec{u}, \vec{v}, v^i) = \frac{R_{ijkl} U^i V^j U^k V^l}{G_{ijkl} U^i V^j U^k V^l}$$

where $G_{ijkl} = g_{jk} g_{il} - g_{ik} g_{jl}$

This curvature can be associated to Gaussian curvature of the surface.

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Question: For the metric of spherical coordinates,

$ds^2 = (dx^1)^2 + (x^1 dx^2)^2 + (x^1 \sin x^2 dx^3)^2$

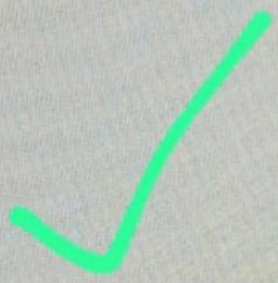
list and calculate the non-zero components R_{ijkl}, if any.

Solution: There are $\frac{n^2(n^2-1)}{12} = \frac{9(8)}{12} = 6$

$\Rightarrow$ There are six non-zero components:

A) R_{1212}, R_{1313}, R_{2323}

B) R_{1213}, R_{1232}, R_{1323}



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$$R^1_{212} = \frac{\partial \Gamma^1_{22}}{\partial x^1} - \frac{\partial \Gamma^1_{21}}{\partial x^2} + \Gamma^1_{23}\Gamma^3_{21} - \Gamma^1_{21}\Gamma^1_{12}$$

$$= \frac{\partial}{\partial x^1}(-x^1) - \Gamma^2_{21}\Gamma^1_{22} = (-1) - \frac{1}{x^1}(-x^1)$$

$$R^1_{212} = -1 + 1 = 0$$

$$R^1_{313} = \frac{\partial \Gamma^1_{33}}{\partial x^1} - \frac{\partial \Gamma^1_{31}}{\partial x^3} + \Gamma^1_{32}\Gamma^2_{31} - \Gamma^1_{31}\Gamma^1_{13}$$

$$= -\sin^2\theta - \Gamma^3_{13}\Gamma^1_{33}$$

$$= -\sin^2\theta - \left(\frac{1}{x^1}\right)(-x^1\sin^2\theta) = 0$$

Similarly:

$R^2_{323} = 0$

$R^1_{213} = 0$

$R^1_{232} = 0$

$R^1_{323} = 0$

Therefore $R_{ijkl} = 0$ for all i,j,k,l.

Since $R_{ijkl} = g_{si} R^s_{jkl}$, so we find the mixed components.

The non-zero christoffel symbols are:

$\Gamma^1_{22} = -x^1$, $\Gamma^1_{33} = -x^1\sin^2\theta$

$\Gamma^2_{12} = \Gamma^2_{21} = \frac{1}{x^1}$, $\Gamma^2_{33} = -\sin\theta\cos\theta$

$\Gamma^3_{13} = \Gamma^3_{31} = \frac{1}{x^1}$, $\Gamma^3_{23} = \Gamma^3_{32} = \cot\theta$

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Exercises:

3) Calculate ds^2 for the event pairs:

a) $E_1(5,1,-2,0)$ and $E_2(0,3,1,-3)$

$$ds^2 = (dx^0)^2 - (dx^1)^2 - (dx^2)^2 - (dx^3)^2$$
$$= (0-5)^2 - (3-1)^2 - (1+2)^2 - (-3)^2$$
$$= 25 - 4 - 9 - 9$$
$$ds^2 = 3$$

$\Rightarrow$ Timelike interval

Also determine the interval type.

b) $E_1(7,2,4,4)$ and $E_2(4,1,2,6)$

$$ds^2 = (3)^2 - (1)^2 - (2)^2 - (2)^2$$
$$= 9 - 1 - 4 - 4 = 0$$

$\Rightarrow ds^2 = 0 \Rightarrow$ light like interval

c) $E_1(5,1,3,3)$ and $E_2(2,-1,1,1)$

$$ds^2 = (3)^2 - (2)^2 - (2)^2 - (2)^2$$
$$= 9 - 4 - 4 - 4 = -3$$

$\Rightarrow$ Space like interval

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Question: Show that

a) Simultaneous events have a spacelike interval.

Solution: Simultaneous events means

$$x_1^0 = x_2^0 \Rightarrow dx^0 = 0$$

for $E_1(x_1^i)$, $E_2(x_2^i)$; $i = 0,1,2,3$

$$\Rightarrow ds^2 = (dx^0)^2 - (dx^1)^2 - (dx^2)^2 - (dx^3)$$
$$\Rightarrow ds^2 = 0 - [(dx^1)^2 + (dx^2)^2 + (dx^3)^2] < 0$$

$\Rightarrow$ Spacelike interval

b) Copositional events have a timelike interval

Solution: Copositional events means

$$x_1^i = x_2^i \text{ for } i = 1,2,3$$
$$\Rightarrow dx^i = 0 \text{ for } i = 1,2,3$$

So,

$$ds^2 = (dx^0)^2 - (dx^1)^2 - (dx^2)^2 - (dx^3)^2$$
$$\Rightarrow ds^2 = (dx^0)^2 > 0$$

$\Rightarrow$ timelike interval

Consider:

$$\bar{\Gamma}^{i}_{jk} = \frac{1}{2}\left(-\bar{g}_{ij,k} + \bar{g}_{jk,i} + \bar{g}_{ki,j}\right)$$

now replacing all values of the metric tensor derivative, we have:

$$\bar{\Gamma}^{i}_{jk} = \frac{1}{2}\left[-g_{rs,t} + g_{st,r} + g_{tr,s}\right]\frac{\partial x^r}{\partial \bar{x}^i}\frac{\partial x^s}{\partial \bar{x}^j}\frac{\partial x^t}{\partial \bar{x}^k} + \frac{1}{2}g_{rs}\left[\frac{\partial^2 x^r}{\partial \bar{x}^j \partial \bar{x}^k}\frac{\partial x^s}{\partial \bar{x}^i} + \frac{\partial^2 x^s}{\partial \bar{x}^j \partial \bar{x}^i}\frac{\partial x^r}{\partial \bar{x}^k}\right]$$

$$\Rightarrow \bar{\Gamma}^{i}_{jk} = \Gamma^{s}_{rt}\frac{\partial x^r}{\partial \bar{x}^i}\frac{\partial x^s}{\partial \bar{x}^j}\frac{\partial x^t}{\partial \bar{x}^k} + g_{rs}\frac{\partial^2 x^r}{\partial \bar{x}^j \partial \bar{x}^k}\cdot\frac{\partial x^s}{\partial \bar{x}^i}$$

(by interchanging the dummy indices r and s in last term)

$\Rightarrow$ Christoffel Symbol is not a general tensor.

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Question: Compute the christoffel symbols corresponding to the Euclidean metric for spherical coordinates.

$$[g_{\mu\nu}] = \begin{bmatrix} 1 & 0 & 0 \\ 0 & (x^1)^2 & 0 \\ 0 & 0 & (x^1)^2\sin^2 x^2 \end{bmatrix}$$

Solution: The formula for christoffel symbols is:

$$\Gamma^{i}_{jk} = \frac{1}{2}\left[-g_{ij,k} + g_{jk,i} + g_{ki,j}\right]$$

$$\Gamma_{221} = \frac{1}{2}\left[-g_{22,1} + g_{21,2} + g_{12,2}\right]$$
$$= \frac{1}{2}(-2x^1 + 0 + 0) = -x^1 \quad \text{or} \quad \boxed{\Gamma_{221} = -x^1}$$

$$\Gamma_{212} = \frac{1}{2}\left(-\cancel{g_{21,2}}^{\,0} + g_{22,1} + \cancel{g_{12,2}}^{\,0}\right)$$
$$= \frac{1}{2}(2x^1) = x^1 \quad \text{or} \quad \boxed{\Gamma_{212} = x^1}$$

$$\Gamma_{122} = \frac{1}{2}\left(-\cancel{g_{12,2}}^{\,0} + g_{22,1} + \cancel{g_{21,2}}^{\,0}\right)$$
$$= \frac{1}{2}(x^1) = x^1$$

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Long Question (Asabi)

(1) Explain symmetry properties 1.0

(2) Find Γ^1_{12} Γ^1_{22} and Γ^2_{11} For $g_{ij} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & (x^1)^2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ 5.0 60

(3) if $U^i = (1,000)$ then Find component of $T^{ij}(P + P/c^2)U^iU^j$ 5.0

(4) 5.0 Find Parmetric $x^1 = b$ and $x^2 = \pi/4$ Polar coordinates.

(5) 5.0 Find $\frac{\partial \vec{v}}{\partial r}$ given $\vec{v} = 2re_r + e_\theta$

$e_r = \cos\theta \hat{i} + \sin\theta \hat{j}$

5.0 $e_\theta = -r\sin\theta \hat{i} + r\cos\theta \hat{j}$

(4) ~~For Fast~~ Show that $V_x = \frac{V_{x'} - V}{1 - \frac{VV_{x'}}{c^2}}$

5.0

(6) How is the Normal stress in a material defined?

7) if $U^i = (c, v^x, v_y, v_z)$

What Diagonal element moment energy —

8) For Falt Riemannian Space $K=0$ Prove that $R=0$ For Space

(Asab)

MTH 623

what are the components of the electromagnetic four potential,

what do you mean by Lorentz group?

List all the non-zero components of R_{ijkl} for $n = 4$ of the type R_{abcd}; acbcad.

Find $\frac{\partial \vec{v}}{\partial \theta}$, given $\vec{v}_1 = r\hat{e}_r + \hat{e}_\theta$ & $\hat{e}_r = \cos\theta\hat{i} + \sin\theta\hat{j}$

, $e_\theta = -r\sin\theta\hat{i} + r\cos\theta\hat{j}$.

Prove That $u_i u^i = c^2$.

In a 2-dimensional space, Non-zero component

$R_{1212} = R_{2121} = -R_{1221} = -R_{2112}$, using this Simplifies the expre. of Riemann curvature.

Find

$\frac{\partial \vec{v}}{\partial r}$, for $\vec{v} = x\hat{e}_r + y\hat{e}_\theta + z\hat{e}_z$ &

$\hat{e}_r = \cos\theta\hat{i} + \sin\theta\hat{j}$, $\hat{e}_\theta = -r\sin\theta\hat{i} + r\cos\theta\hat{j}$

, $e_z = k\hat{i}$ in cylindrical coordinates.

Explain Cauchy's 1st law of motion.

write two component of the electromagnetic four-potential

what do you mean by Lorentz Group?

Math 623

Find $\nabla\varphi$ for $\varphi = \frac{1}{2} x^1 x^2$ in Polar coordinates. (3)

Prove that covariant derivative of Kronecker delta is zero (3)

write the mathematical form of Gauss' Law of Magnetism (3)

write the expressions of electric & magnetic fields in terms of the vector potential. (3)

List all the non-zero components of R_{ijkl} for $n=4$ of the type R_{abcd}; $a<b<c<d$ (3)

what is the Principle stress matrix? (5)

Show that $g_{\alpha\beta;\gamma} = 0$ (5)

v) If $U^i = (c, v^x, v^y, v^z)$

What Diagonal element moment energy —

B) For Falt Riemannin Space

$K=0$ Prove that $R=0$ For Space-

(Asad)

TCO's

Homeloads

Tostly Theory ---

Quiz

Math 623 Today 12:00 PM

Long Question (Asabi)

(1) Explain symmetry properties 1.0

(2) Find Γ^1_{12} Γ^1_{22} and Γ^2_{11} For $g_{ij} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & (x^1)^2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ 5.0 60

(3) if $U^i = (1,000)$ then Find component of $T^{ij}(P + P/c^2)U^iU^j$ 5.0

(4) 5.0 Find Parmetric $x^1 = b$ and $x^2 = \pi/4$ Polar coordinates.

(5) 5.0 Find $\frac{\partial \vec{v}}{\partial r}$ given $\vec{v} = 2re_r + e_\theta$

$e_r = \cos\theta \hat{i} + \sin\theta \hat{j}$

5.0 $e_\theta = -r\sin\theta \hat{i} + r\cos\theta \hat{j}$

(4) ~~For Fast~~ Show that $V_x = \frac{V_{x'} - V}{1 - \frac{VV_{x'}}{c^2}}$

5.0

(6) How is the Normal stress in a material defined?

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The Components of Energy Momentum Tensor of Dust in Minkowsk spacetime (Flat space)

Here we discuss two cases: when frame is comoving and non comoving.

Case I:

A frame moving with the flow of the dust fluid is called comoving frame.

For a co-moving observer, the four velocity reduces to

$$u^i = (c, 0, 0, 0)$$

In this case, the stress-energy tensor takes on the remarkably simple form

$T^{ij} = \rho u^i u^j \rightarrow T^{00} = \rho u^0 u^0 = \rho c^2$

$$T^{ij} = \begin{bmatrix} c^2\rho & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$



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Now in a comoving frame: $V = 0$

$$\Rightarrow \gamma = \frac{1}{\sqrt{1 - v^2/c^2}} = 1$$

$$\Rightarrow u^i = (c, 0, 0, 0)$$

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Here u^i is the proper velocity.

So Recall

$$v^i = \frac{dx^i}{dt} = \frac{d}{dt}(ct, x, y, z) = (c, v^x, v^y, v^z)$$

and $$\frac{d\tau}{dt} = \sqrt{1 - v^2/c^2} \Rightarrow \frac{dt}{d\tau} = \frac{1}{\sqrt{1 - v^2/c^2}} = \gamma$$

Implies proper velocity is:

$$u^i = \frac{d}{d\tau}(ct, x, y, z) = \frac{dt}{d\tau} \cdot \frac{d}{dt}(ct, x, y, z).$$

$$= \frac{dt}{d\tau}(c, v^x, v^y, v^z)$$

$$u^i = \gamma(c, v^x, v^y, v^z)$$

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7) if $U^i = (c, v^x, v_y, v_z)$

What Diagonal element moment energy —

8) For Falt Riemannian Space $K=0$ Prove that $R=0$ For Space

(Asab)